

PROOFS

Subtitle

E. Zimudzi

Department of Computer Science
University of Botswana

16 February, 2026

OUTLINE

1. DIRECT PROOF
2. PROOF BY CONTRAPOSITION
3. PROOF BY CONTRADICTION
4. PROOF BY CASES

- **Theorem:** A mathematical statement that can be shown to be true.
 - Theorem is usually reserved for a statement that is considered at least somewhat important.
 - Less important theorems sometimes are called propositions.
- **Axiom (or postulate):** A statement that is assumed to be true.
- **Lemma:** A less important theorem that is helpful in the proof of other results.
- **Corollary:** A theorem that can be established directly from a theorem that has been proved.
- **Conjecture:** A statement that is being proposed to be a true statement, usually on the basis of some partial evidence.
- **Proof:** A valid argument that establishes the truth of a Theorem.
 - The statements used in a proof can include axioms, definitions, the premises, if any, of the theorem, and previously proven theorems and uses rules of inference to draw conclusions.

Definition

- The integer n is even if there exists an integer k such that $n = 2k$.
- The integer is odd if there exists an integer k such that $n = 2k + 1$.
- Two integers have the **same parity** when both are even or both are odd; they have **opposite parity** when one is even and the other is odd.

Fact: Suppose $a, b \in \mathbb{Z}$. Then,

- $a + b \in \mathbb{Z}$.
- $a - b \in \mathbb{Z}$.
- $ab \in \mathbb{Z}$.

Direct Proof

A direct proof of a conditional statement $p \rightarrow q$ is constructed when the first step is the assumption that the antecedent p is true.

Subsequent steps are constructed using rules of inference, axioms, definitions and/or previously proven theorems with the final step showing that the consequent q must also be true.

Outline for the Direct Proof

Proof: Suppose p

...

...

...

Therefore q

It is generally a good idea to write the first and last line of your proof first, then fill in the gaps.

Example

Prove “If n is an odd integer, then n^2 is odd.”

Proposition “If n is an odd integer, then n^2 is odd.”

Proof Suppose n is odd

Then $n = 2k + 1$ for some $k \in \mathbb{Z}$ definition of odd integer

Thus

$$\begin{aligned} n^2 &= (2k + 1)^2 \\ &= 4k^2 + 4k + 1 \\ &= 2(2k^2 + 2k) + 1 \\ &= 2l + 1 \quad \text{where } l = 2k^2 + 2k, \text{ and } l \in \mathbb{Z} \end{aligned}$$

We can conclude that n^2 is an odd integer.

Consequently, we have proved that if n is an odd integer, then n^2 is an odd integer.

Example

Prove that the sum of two even numbers is even.

Solution

This could be rephrased as:

If x, y are even, then $x + y$ is even.

Assume x and y are even.

$x = 2m, m \in \mathbb{Z}$ by definition of even number

$y = 2n, n \in \mathbb{Z}$ by definition of even number

$$\begin{aligned}x + y &= 2m + 2n \\&= 2(m + n) \\&= 2r \quad \text{where } r = m + n, \text{ and } r \in \mathbb{Z}\end{aligned}$$

That is, $x + y = 2r$ is an even number. We have proved that the sum of two even numbers is even.

Proof by Contraposition

A type of **indirect proof**.

Makes use of the fact that the conditional statement $p \rightarrow q$ is equivalent to its contrapositive, $\neg q \rightarrow \neg p$.

We take $\neg q$ as a premise, and show that $\neg p$ must follow.

Outline for Contrapositive Proof

Proposition: if p , then q .

Proof: Suppose $\neg q$

...

...

...

Therefore $\neg p$

Example

Prove that if $7x + 9$ is even, then x is odd.

Proposition Suppose $x \in \mathbb{Z}$. If $7x + 9$ is even, then x is odd.

Proof. Suppose x is not odd

Thus x is even, so $x = 2k$ for some integer k .

Then,

$$\begin{aligned} 7x + 9 &= 7(2k) + 9 \\ &= 14k + 8 + 1 \\ &= 2(7k + 4) + 1 \\ &= 2r + 1 \quad \text{where } r = 7k + 4, \text{ and } r \in \mathbb{Z} \end{aligned}$$

This tells us that $7x + 9$ is odd.

Therefore $7x + 9$ is not even.

The negation of the conclusion of the conditional statement implies that the hypothesis is false, the original conditional statement is true.

$\therefore$ We have proved that “If $7x + 9$ is even, then n is odd.”

Example

Prove that if n is an integer and $3n + 2$ is odd, then n is odd.

Solution

Assume that n is even, i.e. $n = 2k$, for some integer k , by definition of an even integer. Substituting $2k$ for n , we find that

$$\begin{aligned} 3n + 2 &= 3(2k) + 2 \\ &= 6k + 2 \\ &= 2(3k + 1) \\ &= 2r \quad \text{where } r = 3k + 1, \text{ and } r \in \mathbb{Z} \end{aligned}$$

This tells us that $3n + 2$ is even. This is the negation of the premise of the theorem. Because the negation of the conclusion of the conditional statement implies that the hypothesis is false, the original conditional statement is true.
 $\therefore$ We have proved that “If $3n + 2$ is odd, then n is odd.”

Example

Suppose $x \in \mathbb{Z}$. If $x^2 - 6x + 5$ is even, then x is odd.

Proof Suppose x is not odd

Thus x is even, so $x = 2k$ for some integer k .

So

$$\begin{aligned}x^2 - 6x + 5 &= (2k)^2 - 6(2k) + 5 \\&= 4k^2 - 12k + 4 + 1 \\&= 2(2k^2 - 6k + 2) + 1 \\&= 2r + 1 \qquad \text{where } r = 2k^2 - 6k + 2, \text{ and } r \in \mathbb{Z}\end{aligned}$$

Consequently $x^2 - 6x + 5$ is odd.

Therefore $x^2 - 6x + 5$ is not even.

The negation of the conclusion of the conditional statement implies that the hypothesis is false, the original conditional statement is true.

$\therefore$ We have proved that “If $x^2 - 6x + 5$ is even, then n is odd.”

Example

Prove that for all integers a and b , if $a + b$ is odd, then a is odd or b is odd.

The contrapositive of the statement we are trying to prove is: for all integers a and b , if a and b are even, then $a + b$ is even (Note that our assumption that a and b are even is the negation of a or b is odd. We used De Morgan's law here $\neg(a \vee b) \equiv (a \wedge b)$).

Proof

Let a and b be integers. Assume that a and b are even.

Then $a = 2k$ and $b = 2l$ for some integers k and l .

Now $a + b = 2k + 2l = 2(k + l)$.

Since $k + l$ is an integer, we see that $a + b$ is even, completing the proof.

Definition

The real number r is rational if there exist integers p and q with $q \neq 0$ such that $r = p/q$.
A real number that is not rational is called irrational.

Proof by Contradiction

Not limited to proving just conditional statements, can be used to prove any kind of statement.

First assume that a proposition is not true, then through that premise and logic find a contradiction that shows the original premise must have been incorrect, and therefore, the proposition was true.

For one proposition p , assume $\neg p$ is true, then find a contradiction that shows $\neg p$ is false, so p is true.

For an implication, $p \rightarrow q$, assume p and $\neg q$ are true, then find a contradiction that shows either $p \rightarrow q$ or $\neg q \rightarrow \neg p$.

Proof by Contradiction with One Proposition

Example Proof that $\sqrt{2}$ is irrational.

Solution

Let p be the proposition “ $\sqrt{2}$ is irrational.”

We suppose that $\neg p$ is true.

$\neg p$ is the statement “It is not the case that $\sqrt{2}$ is irrational, i.e. $\sqrt{2}$ is rational.

We show that assuming $\neg p$ is true leads to a contradiction.

Assume $\sqrt{2}$ is rational.

Then, there exists 2 integers a and b such that $\sqrt{2} = \frac{a}{b}$, $b \neq 0$, and a and b have no common factors.

$$\sqrt{2} = \frac{a}{b}$$

$$2 = \frac{a^2}{b^2}$$

$$2b^2 = a^2$$

Cont.

Therefore a must be even. Thus, as we know that a and b are not both even, it follows that b is odd. So a can be written as $a = 2c$, for some $c \in \mathbb{Z}$

$$2b^2 = (2c)^2$$

$$2b^2 = 4c^2$$

$$b^2 = 2c^2$$

Therefore b must be even. But previously, we deduced that b is odd. Thus we have a contradiction b is even, and b is odd.

Since a and b are both even, they have a common factor.

Therefore $\sqrt{2}$ is irrational.

Practice

Proof by contradiction “If n is an integer and $3n + 2$ is even, then n is even.”

Solution

Let p : $3n + 2$ is even, and q : n is even.

Assume p and $\neg q$, i.e. $3n + 2$ is even, $n \in \mathbb{Z}$, and n is odd.

We suppose that $\neg q$ is true.

Since $3n + 2$ is even, then $3n$ is even by subtraction of 2 even integers. Then $3n - n$ must be odd by subtraction of an even and odd integer. However, $3n - n = 2n$, which is even by definition of an even number, which is a contradiction of our original assumptions.

$\therefore$ “If n is an integer and $3n + 2$ is even, then n is even.”

Proving Conditional Statements by Contradiction

Example

Prove that “If $n^3 + 5$ is odd, then n is even.”

Solution

Let p : If $n^3 + 5$ is odd, and p : n is even.

Assume p and $\neg q$, i.e. $n^3 + 5$ is odd, $n \in \mathbb{Z}$, and n is odd. $\therefore n = 2k + 1$ for $k \in \mathbb{Z}$

We show that the above assumption leads to a contradiction.

$$\begin{aligned}n^3 + 5 &= (2k + 1)^3 + 5 \\&= 8k^3 + 12k^2 + 6k + 1 + 5 \\&= 8k^3 + 12k^2 + 6k + 6 \\&= 2(4k^3 + 6k^2 + 3k + 3) \\&= 2t \qquad t = 4k^3 + 6k^2 + 3k + 3, t \in \mathbb{Z}\end{aligned}$$

Therefore $n^3 + 5 = 2t$, i.e. $n^3 + 5$ is even. This contradicts the statement that $n^3 + 5$ is odd.

Hence, proven that “If $n^3 + 5$ is odd, then n is even.”

Proving by Contradiction

Example

Prove: There are no integers x and y such that $x^2 = 4y + 2$.

Proof

We proceed by contradiction.

Suppose there are integers x and y such that $x^2 = 4y + 2 = 2(2y + 1)$, that is $x^2 = 2(2y + 1)$, so x^2 is even, meaning that x is even.

So $x = 2k$ for some integer k . Then $x^2 = (2k)^2 = 4k^2$.

This in turn gives $2k^2 = (2y + 1)$. But $2k^2$ is even, and $2y + 1$ is odd, so these cannot be equal. Thus we have a contradiction, so there must not be any integers x and y such that $x^2 = 4y + 2$.

Proof by Cases

Used to prove a theorem by considering different cases separately.

The validity of a proof by cases rests on the equivalence:

$$[(p_1 \vee p_2 \vee \dots \vee p_n) \rightarrow q] \leftrightarrow [(p_1 \rightarrow q) \vee \dots (p_n \rightarrow q)].$$

The result must be true for all cases. If it fails for one case, then the result is false.

Typical cases

- a) Case 1: n is even
Case 2: n is odd
- b) Case 1: $n > 0$
Case 2: $n = 0$
Case 3: $n < 0$

e.g. Prove that if $n \in \mathbb{Z}$, then $n^2 + 3n + 5$ is an odd integer.

Proof We proceed by cases, according to whether n is even or odd.

case 1, n is even, then $n = 2k$, $k \in \mathbb{Z}$

So,

$$\begin{aligned}n^2 + 3n + 5 &= 4k^2 + 6k + 5 \\&= 2(2k^2 + 3k + 2) + 1 \\&= 2r + 1 \qquad r = 2k^2 + 3k + 2, r \in \mathbb{Z}\end{aligned}$$

case 2, n is odd, then $n = 2k + 1$, $k \in \mathbb{Z}$

$$\begin{aligned}n^2 + 3n + 5 &= 4k^2 + 4k + 1 + 6k + 3 + 5 \\&= 2(2k^2 + 5k + 4) + 1 \\&= 2r + 1 \qquad r = 2k^2 + 5k + 4, r \in \mathbb{Z}\end{aligned}$$

Thus, $n^2 + 3n + 5$ is an odd integer.

Biconditionals

In order to establish the truth of the statement $p \leftrightarrow q$, use the fact that $(p \leftrightarrow q)$ is equivalent to $(p \rightarrow q) \wedge (q \rightarrow p)$ and prove both implications using any of the methods of proof.

e.g. Prove: For any integer n , n is odd if and only if n^2 is odd.

Proof

In order to prove this statement, we must prove two implications:

- If n is odd, then n^2 is odd.
- If n^2 is odd, then n is odd.

Proof of (a): We give a direct proof of this statement. Assume n is an odd integer. Then $n = 2k + 1$ for some integer k .

Then $n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$, which is twice an integer plus 1. Thus, n^2 is odd.

Proof of (b): We give a proof of the contrapositive of this statement:

“If n is even (not odd), then n^2 is even (not odd)”.

Assume n is an even integer.

Then $n = 2k$ for some integer k .

Then $n^2 = (2k)^2 = 4k^2 = 2(2k^2)$, which is an even integer.

∴ Proven.

e.g. Prove that $m^2 = n^2$ if and only if $m = n$ or $m = -n$

We want to prove that

- $\forall m, n \in \mathbb{R}, (m^2 = n^2) \rightarrow m = n, \text{ or } m = -n.$
- $\forall m, n \in \mathbb{R}, m = n, \text{ or } m = -n \rightarrow (m^2 = n^2).$

Proof

First, we prove that if $m^2 = n^2$, then $m = n$, or $m = -n$. Assume $m^2 = n^2$, then $m^2 - n^2 = 0$. Since $m^2 - n^2 = (m = n)(m = -n)$, we have $(m = n)(m = -n) = 0$. This implies that $m = n = 0$, or $m = -n = 0$ which is equivalent to $m = n$ or $m = -n$.

Next we prove that if $m = n$ or $m = -n$ then $m^2 = n^2$. Assume $m = n$, or $m = -n$.

If $m = n$, we can square both sides of the equation to obtain $m^2 = n^2$.

If $m = -n$, we can square both sides of the equation to obtain $m^2 = n^2$.

Hence, if $m = n$, or $m = -n$, then $m^2 = n^2$.

The End